The potential use of saliva to detect recurrence of disease in women with breast carcinoma

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Abstract

Background: Approximately 1 woman in every 10 will develop breast cancer in her lifetime. It has been shown that screening for breast cancer can reduce breast cancer mortality. The use of a saliva-based test could prove to be very useful in post-operative and/or adjunctive therapy management of breast cancer patients.

Methods: The following study was undertaken to establish the possible usefulness of the salivary protein product of the oncogene c-*erbB-2* in following patients diagnosed with carcinoma of the breast. Included in this study were 25 patients with a mean age of 54 years with varying histological diagnoses and stages of carcinoma of the breast. ELISA assays for c-*erb*B-2 and CA 15–3 were performed on serum and stimulated whole saliva samples collected on all patients prior to any adjunct therapy or surgery and sequentially during therapy.

Results: The results of the GLM analyses using marker concentration as the dependent variable and treatment regimen and the serial assessments as independent variables yielded a significant overall model for both the serum ($P < 0.007$) and salivary ($P < 0.017$) c-*erb*B-2 markers. The model for serum c-*erb*B-2, however, exhibited a significant difference for treatment regimen ($P < 0.001$) with the chemotherapy and radiation treatment regimen being significantly different ($P < 0.001$) from the other treatment therapies. Time (serial assessments) was not significant. The model for the salivary c-*erb*B-2 marker was reversed. Treatment regimen was not significant for this model; however, time (serial assessments) was significant ($P < 0.002$). The serum and salivary CA 15–3 marker models yielded no significant results. Paired *t*-test analyses indicated that only the salivary c-*erb*B-2 concentrations exhibited a significant difference between the pre- and post-therapy values ($t = 4.245$, $P < 0.0001$). Additionally, salivary c-erbB-2 displayed greater percent reductions across all therapies as compared to the other markers.

Conclusions: This preliminary study appears to indicate that c-*erb*B-2 protein expression in saliva may be a very useful diagnostic tool for measuring patient response to chemotherapy and/or surgical treatment of their disease.

Key words: breast cancer; c-*erb*B-2; recurrence; saliva

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Cancer is the second leading cause of death in the United States. An estimated 44,000 women will die of breast cancer, while

150,000 new female cases will be diagnosed this year (1–3) alone. Approximately 1 woman in every 10 will develop breast cancer in her lifetime.

Regrettably, many deaths due to cancer are premature. It has been shown that screening for breast cancer can reduce breast cancer mortality (4). Among women aged 50 and older, studies have demonstrated a 20–40% reduction in breast cancer mortality for women screened by mammography and clinical breast examination (1–8). However, among women between 40 and 49 years of age, the mortality rate is reduced only 13–23%. Despite the benefits from screening, these services are far from being fully utilized (5, 6). For example, women aged 50 and older become less compliant and less likely to continue with the 18–24-month periodic examinations (7). Reasons for the decrease in compliance among women 50 years of age and older are speculative, but may be due in part to the high cost of the screening procedures and the lack of available mammography services in some areas of the nation (8).

While physical examination and mammography are useful screening procedures for the early detection of breast cancer, they are also labor intensive and require health professionals who are highly trained and experienced. Despite efforts to provide accurate diagnoses, these screening procedures can produce a substantial percentage of false positive and false negative results, especially in women with dense parenchymal breast tissue (1–8).

The use of saliva as an adjunct diagnostic test has advantages and may supplement current methodologies (9). It has been determined that a number of tumor markers are present in saliva (10–15); therefore, its use as a diagnostic fluid could have significant diagnostic and logistical advantages when compared to serum. From a logistical perspective, the collection of saliva is safe (i.e. no needle punctures), non-invasive and relatively simple, and may be collected repeatedly without discomfort to the patient (15).

A saliva-based test would also be extremely useful in the initial management of cancer patients. An inexpensive saliva test used in conjunction with a mammograph may increase the overall diagnostic value of the mammograph and reduce the number of false positives and negatives currently associated with the test (2, 5, 16). Additionally, if a strong relationship between the presence of breast cancer and the salivary markers were proven, the diagnostic potential could reach many individuals who, for personal, logistical or economical reasons, lack access to preventive care. More importantly, the diagnosis of breast cancer at an earlier stage allows a woman more choice in selection of various treatment options.

Saliva also could be a cost-effective method for monitoring the efficiency of chemotherapy and/or other post-operative care. Individuals should experience decreases in marker concentrations if the treatment regimen is effective. The ability to economically and logistically provide an adjunct diagnostic test for mammography, and for physician and self-breast examination may reduce morbidity and mortality rates for breast cancer and thereby reduce overall national health care expenditures. There is a definitive need and use for a simple, safe, economic adjunct diagnostic test for the detection and post-operative follow-up of patients afflicted with carcinoma of the breast (8, 16). The literature has also indicated that saliva, due to its logistical and biochemical advantages, would be an excellent medium to use as a diagnostic tool (9–15).

c-*erb*B-2, also known as Her-2/neu, is a prognostic breast cancer marker assayed in tissue biopsies from women diagnosed with malignant tumors (17, 18). c-*erb*B-2 protein is over-expressed in approximately 20–30% of malignant breast tumors and has been used in post-operative follow-up evaluation as an indicator of patient relapse (19, 20). Additionally, circulating c-*erb*B-2 receptor protein levels have also been detected in serum (20–24). Studies investigating the serological levels of c-*erb*B-2 among women with carcinoma of the breast have shown that the oncoprotein is elevated in 9–52% of the patients, and additionally increases in the presence of regional and distant metastases (17–24).

Recognizing the diagnostic potential of the c-*erbB-2* marker, the investigators initiated this study to determine if salivary levels of c-*erb*B-2 protein could be useful as a marker for post-operative monitoring for patients diagnosed with carcinoma of the breast. Consequently, the study was conducted to compare serial assessments of salivary and serological c-*erb*B-2 and CA 15–3 concentrations among women. CA 15–3 was included in this study because of its continued use as a marker of post-operative follow-up evaluation and as an indicator of patient relapse (25).

Materials and methods

Patient characteristics

The patients in this study were 25 women with varying stages of carcinoma of the breast as defined by TNM system proposed by the American Joint Commission on Cancer Staging and End Results Reporting (26). The women were consecutively sampled as they entered the University of Mississippi Medical Center Oncology Clinic upon referral for treatment. After agreeing to and signing the Institutional Review Board-approved consent form, serum and saliva samples were collected first at the initial visit (baseline), then serially as the patient returned to clinic for post-operative follow-up. Due to the nature of the outpatient clinic where varying treatment regimens, treatment compliance, access to care, and travel economics affect the regularity of return visits, regular sampling intervals could not be established. The women, however, were followed throughout their treatment protocols to further ascertain whether salivary c-*erb*B-2 protein concentrations would be predictive for recurrence of disease. Clinical cancer variables (e.g. receptor status, staging, etc.) were obtained from the patient's pathology reports.

Variables concerning c-*erb*B-2 protein concentrations and its relationship to race, age, weight, other systemic illnesses (e.g. diabetes), the use of prescription medications, salivary flow rates, and the presence of periodontal disease have been published in previous studies and have no effect on salivary c-*erb*B-2 concentrations (27–29). None of the patients in this study showed evidence of negative oral pathology, which might confound this study.

Stimulated whole saliva collection

The participants were seen between 8.00 AM and 5.00 PM. The study required that the individuals not eat, drink, smoke, or brush their teeth for at least 60 min prior to saliva collection. The subject first swallowed accumulated saliva in the mouth. An unflavored, unsweetened piece of chewing gum base was placed in the mouth and masticated (60 chews/min) as monitored with a metronome (30). Accumulated saliva was expectorated into a pre-weighed cup for a total of 5 min. Flow rates (ml/min) were determined gravimetrically and the physical characteristics were recorded (e.g. blood in saliva). Specimens containing tinges of blood were discarded. Collected specimens were immediately frozen until they were ready for analysis.

SWS is the method of choice in most epidemiological studies dealing with biomarkers because of the large amount of saliva that is produced, the ease in collection, and the reproducibility in saliva production (29). More importantly, it is not subjected to circadian rhythms; hence, the total volume of stimulated whole saliva is independent of time of day making it a suitable method for collecting saliva during the 8.00 AM to 5.00 PM time period (31, 32). Additionally, salivary c-*erb*B-2 concentrations are not flow rate dependent (11, 29). Blood was drawn immediately after the SWS specimen was collected.

Laboratory methods

The frozen SWS and serum samples were thawed and clarified by centrifugation (500–1500 g for 20 min) to separate the soluble c-*erb*B-2 materials from particulates in the saliva specimens. The clarified SWS were assayed using a kit (Pierce Chemical Company, USA) for protein concentration using bicinchoninic acid procedure (33). For this, 0.1 ml SWS was placed in microtiter plates and assay reagents added. After incubation at 37°C for 30 min, protein levels were quantified using a spectrophotometer set at 562 nm. The final concentration was determined from a standard curve established from the serially diluted bovine albumin standards.

C-*erb*B-2 enzyme-linked immunosorbent assay (ELISA)

The saliva and sera of the patients with breast cancer were assayed by enzyme-linked immunosorbent assay (Oncogene™ Research Products, USA) for antigens directed against the extracellular domain of the c-*erb*B-2 receptor (23, 34). The assays are a two-site solid phase enzyme immunoassay. The molecules of the antigens of interest are 'sandwiched' between two monoclonal antibodies, the first one attached to the ELISA solid phase and the second one linked to the horseradish peroxidase (enzymatic conjugate). After washing, the enzymatic reaction develops a color proportional to the amount of c-*erb*B-2 antigens present in the assay. The absorbance was read at 490 nm using a spectrophotometer and concentrations are calculated from standard curves constructed from known concentrations of the ligand. Samples from two healthy individuals that have been followed for several years were also assayed on each microtiter plate in order to assure for batch intra- and inter-reliability (29).

The precision (coefficient of variance) of the c-*erb*B-2 assay is between 7.9 and 8.7% with a detection sensitivity of 10 human Neu units/ml. A detailed description concerning the reliability of the assay using saliva can be found in Streckfus (11, 29) and information related to possible interferences is reviewed in Payne (23) and Schwartz (34).

CA 15–3 enzyme immunoassay (EIA)

CA 15–3 assays were determined by using EIA kits (CIS Bio International, France). The CA 15–3 assay is a two-site solid phase enzyme immunoassay (35). The molecules of CA 15–3 are 'sandwiched' between two monoclonal antibodies. The first one is attached to the ELISA solid phase and the second one linked to the horseradish peroxidase (enzymatic conjugate). After washing, the enzymatic reaction develops a color proportional to the amount of CA 15–3 present in the assay. Absorbance is read at 490 nm (horseradish peroxidase) using a spectrophotometer and the concentration is calculated from a standard curve constructed from known concentrations of the ligand. The CA 15–3 assay is designed to assay serum specimens. Saliva supernatants were substituted in place of the serum for salivary CA 15–3 determinations. CA 15–3 concentrations were expressed as units/ml.

Statistical analyses

Statistical analyses were performed using the SPSS^{TM} statistical software package (36). The data were analyzed from several different perspectives. Initially, descriptive analyses were conducted for mean values, cross-tabulations and graphical representations of the data. For the demographic and tumor data obtained from the questionnaire and pathology reports, cross-tabulations were constructed. Chi-square and Fisher exact probability tests were used to analyze the dichotomous variables while the Kendall's Tau was used to assess ordinal data. Mean values for demographic, clinical and pathologic variables were determined for c-*erb*B-2 concentrations; however, due to the small number in the study and the lack of normal distribution of tumor stages across the variables, the determinations yielded little information and/or could be misleading and, consequently, are not reported in this study. However, information concerning c-*erb*B-2 concentrations and the aforementioned variables can be found in previous studies performed by the authors (11, 27–29).

Receiver operating characteristic (ROC) analyses to investigate the appropriate cutoff value for each biomarker were determined using a combination of patients who accepted to be in the longitudinal study and from those subjects who refused continuation in the study and/or those who dropped out of the study shortly after baseline values were determined. Separately for each marker their concentrations were recoded into dichotomous variables using various percentile cutoff values from the pooled control and cancer groups. Two-by-two tables were used to compute the sensitivity and specificity values of each biomarker for detecting disease for each cutoff value. ROC curves (sensitivity vs. 1-specificity) were constructed for c-*erb*B-2 and CA 15–3 concentrations in both saliva and serum. The optimum cutoff value for each marker was determined by using the cutoff value that produced the largest percentage of area under its ROC curve (33–38).

As previously mentioned, serial assessments for each marker across the patient population were not uniform with respect to the interval of time each sample was collected. Consequently, the marker assessments for the individual subjects were grouped in 180-day intervals yielding five time periods that were suitable for analyses. Marker values were then recoded into two groups: those above (positive) and below (negative) the marker cutoff value. Kendal's Tau was used to assess significant positive and negative frequency differences across the 180-day intervals.

Serial data were analyzed using the general linear model (GLM) procedure. We utilized the intrinsic continuity of the repeated measures factor by focusing on the nature of the longitudinal data taking into account our sample size. The method uses orthogonal polynomial coefficients to model the response curves (40, 41). These methods provide data reduction from several time points to a few biologically meaningful summary statistics and eliminate discussions of biologically insignificant changes in outcomes and reduce multiple testing problems (e.g. the multiplicity phenomena). The polynomials are easy to interpret and are appropriate for all sample sizes including those too small to sustain an appropriate multivariate analysis (40, 41).

Subjects receiving radiation and chemotherapy were also evaluated with respect to their therapies and their relationship to serum and salivary marker levels. Pre- and post-treatment marker levels were compared using the paired samples *t*-test. Percentages for the various treatments were also determined using pre-therapy levels as the denominator.

Results

The study group consisted of 25 women with a mean age of 53.6 ($\pm$10.8) ranging from 33 to 75 years of age. The group was composed of 12 Caucasians, 13 African–Americans, 9 tobacco users, 16 non-tobacco users, 9 pre-menopausal women, and 16 post-menopausal women. The mean number of days of follow-up observation was 236 with a maximum limit of 1230 days.

The women with cancerous lesions were diagnosed with ductal carcinoma *in situ* ($n = 2$), infiltrating ductal carcinoma ($n = 19$), and four miscellaneous lesions consisting of infiltrating lobular carcinoma ($n = 3$), and one colloid carcinoma ($n = 1$).

With respect to the staging of the cancer tumors, there were two Stage 0 (TisN0M0), four Stage I composed of one (T1aN0M0) and one (T1bN0M0) and two (T1N0M0), five Stage IIA composed of five (T2N0M0), four Stage IIB composed of four T2N1M0, four Stage IIIA composed of four T3N1M0 and three Stage IV composed of one (T1cN1M1) and two (T2N1M1). Three were not staged at the time of this report. As illustrated previously, 11 subjects with carcinoma of the breast were node positive and 14 were node negative. Three individuals were diagnosed with evidence of distant metastases.

With respect to Elston grading, four women were assessed as '1' or low-grade, four were assessed as '2' or moderate-grade, and 10 as '3' or high-grade tumors. Seven were without assessments. Ninety percent of the high-grade lesions were found in tumors graded Stage IIa and above. The Kendal's Tau probability test revealed no significant relationship between tumor grading and ER/PR and Her-2/neu receptor status.

Estrogen (ER) and progesterone (PR) receptor status revealed that seven were ER negative while 13 were ER positive. Five individuals were not rated. The PR status revealed that 10 were PR negative and 10 were PR positive. Likewise, five individuals were not rated. Fisher exact probability test revealed no significant association between ER and PR status.

Her-2/neu (Her-2) receptor status determined that 10 were Her-2 negative and 4 were Her-2 positive. Eleven individuals were not assessed for Her-2 status as this was not a routine procedure at the beginning of this study. Fisher exact probability test revealed no significant relationship between Her-2 status and ER/PR status.

Within the group of 25 patients, five received surgical intervention, eight received chemotherapy, four received radiation treatment and eight had a combination of radiation and chemotherapy. Chemotherapy regimens varied among the 16 patients receiving this mode of treatment. Four received cyclophosphamide (C) methotrexate (M) fluorouracil (F) chemotherapy, five received doxorubicin (D) and paclitaxel (P), one received CDF, two received CMF with D and P, three received CMF with P, and one was treated solely on tamoxifen citrate.

The comparison of receiver operator curves (fourth level) produced a cutoff value of 80 units/ml for salivary and 2500 units/ml for serum c-*erb*B-2 concentrations. Serum CA 15–3 determinations yielded a cutoff value of 20 units/ml while salivary CA 15–3 determinations yielded a 5.0 units/ml cutoff value.

Using the aforementioned cutoff values, salivary and serum c-*erb*B-2 concentrations were able to detect 87 and 94% of the subjects with cancer, respectively. This compares to 62 and 75% for the salivary and serum CA 15–3 marker. These values correspond to previous values reported by the investigators of this study (11).

Mean values for serial determinations of serum c-*erb*B-2, salivary c-*erbB-2*, serum CA 15–3, and salivary CA 15–3 concentrations over time (days from baseline) are illustrated in Table 1. The results are stratified according to treatment regimen. Total values are also presented.

The results of the GLM analyses (Table 2) using marker concentration as the dependent variable and treatment regimen and the serial assessments as independent variables yielded a significant overall model for both the serum ($P < 0.007$) and salivary ($P < 0.017$) c-*erb*B-2 markers. The model for serum c-*erb*B-2, however, exhibited a significant difference for treatment regimen ($P < 0.001$) with the chemotherapy and radiation treatment regimen being significantly different ($P < 0.001$) from the other treatment therapies. Time (serial assessments) was not significant. The model for the salivary c-*erb*B-2 marker was reversed. Treatment regimen was not

Table 1.

Treatment	Marker	Baseline	1–180 days	181–360 days	361–540 days	540+ days
Totals for entire group	Serum c-*erb*B-2	3784.93 (±416.32)	3599.25 (±242.20)	3667.22 (±197.29)	3274.63 (±186.90)	3465.56 (±223.52)
	Saliva c-*erb*B-2	158.34 (±9.23)	121.78 (±4.45)	89.73 (±1.25)	99.23 (±1.72)	67.61 (±4.58)
	Serum CA 15–3	35.42 (±22.49)	33.03 (±11.53)	23.63 (±12.28)	24.59 (±24.40)	24.18 (±11.99)
	Saliva CA 15–3	13.85 (±4.90)	11.26 (±2.45)	16.39 (±5.40)	17.51 (±9.28)	8.16 (±1.85)
Surgery only	Serum c-*erb*B-2	3337.89 (±592.17)	2842.43 (±540.58)	Not available	2817.61 (±592.17)	2997.01 (±540.58)
	Saliva c-*erb*B-2	204.05 (±37.87)	126.19 (±34.57)	192.28 (±84.68)	185.67 (±37.87)	45.60 (±34.57)
	Serum CA 15–3	27.97 (±12.60)	28.07 (±28.07)	Not available	29.87 (±16.27)	18.38 (±16.27)
	Saliva CA 15–3	7.36 (±9.63)	8.55 (±9.64)	0.48 (±0.00)	0.43 (±0.00)	2.58 (±12.44)
Chemotherapy only	Serum c-*erb*B-2	2878.23 (±468.15)	3424.37 (±331.03)	3031.58 (±441.38)	3609.73 (±662.07)	Therapy over
	Saliva c-*erb*B-2	158.86 (±37.87)	96.40 (±19.43)	38.59 (±28.23)	34.24 (±42.34)	Therapy over
	Serum CA 15–3	55.81 (±11.51)	36.07 (±6.84)	25.55 (±12.60)	26.20 (±19.93)	Therapy over
	Saliva CA 15–3	7.82 (±8.80)	8.33 (±5.56)	7.16 (±9.64)	12.05 (±15.24)	Therapy over
Radiation only	Serum c-*erb*B-2	3370.68 (±662.07)	2513.16 (±662.07)	2560.48 (±764.49)	2248.53 (±936.31)	2416.68 (±764.49)
	Saliva c-*erb*B-2	123.02 (±41.35)	104.26 (±42.34)	104.50 (±40.33)	34.53 (±59.88)	23.47 (±48.89)
	Serum CA 15–3	19.05 (±14.09)	19.31 (±13.08)	19.38 (±16.27)	12.35 (±28.18)	32.97 (±9.96)
	Saliva CA 15–3	28.14 (±10.78)	34.39 (±12.44)	27.74 (±10.78)	79.52 (±21.55)	22.78 (±21.55)
Chemotherapy and radiation	Serum c-*erb*B-2	5178.16 (±468.15)	4329.55 (±331.03)	4119.26 (±296.09)	3607.62 (±441.38)	3923.85 (±367.25)
	Saliva c-*erb*B-2	149.61 (±29.94)	152.70 (±20.54)	103.31 (±18.05)	95.32 (±25.53)	85.24 (±21.86)
	Serum CA 15–3	32.97 (±9.96)	35.23 (±7.28)	23.78 (±6.30)	21.91 (±12.60)	30.02 (±12.60)
	Saliva CA 15–3	15.77 (±8.80)	10.47 (±5.56)	17.17 (±4.70)	9.01 (±10.78)	8.47 (±8.15)

Note: Serum and salivary c-*erb*B-2 concentrations are expressed as HNU/ml. Serum and salivary CA 15–3 concentrations are expressed as units/ml.

Table 2.

Source	Type III sum of squares	d.f.	Mean square	*F*	Significance
Serum c-*erb*B-2 Model[a]	68682094.082	18	3815671.893	2.168	0.007
TREAT	48287125.311	3	16095708.437	9.144	0.000
DAY2	4624065.003	4	1156016.251	0.657	0.623
TREAT * DAY2	11765501.338	11	1069591.031	0.608	0.819
Salivary c-*erb*B-2 Model[b]	258962.161	18	14386.787	1.951	0.017
TREAT	45991.914	3	15330.638	2.079	0.106
DAY2	133559.973	4	33389.993	4.527	0.002
TREAT * DAY2	85745.109	11	7795.010	1.057	0.401
Serum CA 15–3 Model[c]	9750.787	18	541.710	0.686	0.816
Salivary CA 15–3 Model[d]	9905.675	18	550.315	1.184	0.291

[a]Serum c-*erb*B-2 Model: $R^2 = 0.242$ (adjusted $R^2 = 0.131$).
[b]Salivary c-*erb*B-2 Model: $R^2 = 0.208$ (adjusted $R^2 = 0.101$).
[c]Serum CA 15–3 Model: $R^2 = 0.116$ (adjusted $R^2 = -0.053$).
[d]Salivary CA 15–3 Model: $R^2 = 0.191$ (adjusted $R^2 = 0.030$).

significant for this model; however, time (serial assessments) was significant ($P < 0.002$). The serum and salivary CA 15–3 marker models yielded no significant results.

The percent of elevated assessments above and below the cut points for serum c-*erb*B-2, salivary c-*erb*B-2, serum CA 15–3, and salivary CA 15–3 concentrations over time (days from baseline) are illustrated in Table 3.

Table 4 represents pre- and post-therapy marker values. Paired *t*-test analyses indicated that only the salivary c-*erb*B-2 concentrations exhibited a significant difference between the pre- and post-therapy values ($t = 4.245$, $P < 0.0001$). Additionally, salivary c-*erb*B-2 displayed greater percent reductions across all therapies as compared to the other markers (Tables 5 & 6).

Table 3.

Marker	Cut point values	180-day cycle (%)				
		Baseline	1–180	181–360	361–540	540+
Serum c-*erb*B-2	Percent above cut point 2500 HNU/ml	80.0	81.0	90.6	75.0	81.8
	Percent below cut point 2500 HNU/ml	20.0	19.0	9.4	25.0	18.2
Salivary c-*erb*B-2						
	Percent above cut point 80 HNU/ml	76.0	60.9	47.2	31.8	33.3
	Percent below cut point 80 HNU/ml	24.0	39.1	52.8	65.2	66.7
Serum CA 15–3	Percent above cut point 20 units/ml	68.0	64.0	63.9	59.1	62.5
	Percent below cut point 20 units/ml	32.0	36.0	36.1	40.9	37.5
Salivary CA 15–3	Percent above cut point 5 units/ml	56.0	62.0	72.2	50.0	58.3
	Percent below cut point 5 units/ml	44.0	38.0	27.8	50.0	41.7

Table 4.

Marker	*n*	Mean	SEM	*t*	*P*
Pre-serum c-*erb*B-2 HNU/ml	36	3634.72	238.33	1.718	0.095
Post-serum c-*erb*B-2 HNU/ml		3384.72	152.24		
Pre-salivary c-*erb*B-2 HNU/ml	32	131.31	12.68	4.245	0.0001
Post-salivary c-*erb*B-2 HNU/ml		84.25	7.32		
Pre-serum CA 15–3 units/ml	32	30.09	4.19	1.830	0.077
Post-serum CA 15–3 units/ml		24.06	2.75		
Pre-salivary CA 15–3 units/ml	30	13.03	3.04	0.630	0.530
Post-salivary CA 15–3 units/ml		11.80	1.70		

Table 5.

Type of therapy	*n*	Percent serum c-*erb*B-2 reduction	Percent salivary c-*erb*B-2 reduction	Percent serum CA 15–3 reduction	Percent saliva CA 15–3 reduction
Radiation	19	3	32	8	No change
Chemotherapy	9	9	23	8	No change
Radiation and chemotherapy	2	No Change	18	No change	No change

Table 6.

Type of chemotherapy	*n*	Percent serum c-*erb*B-2 reduction	Percent salivary c-*erb*B-2 reduction	Percent serum CA 15–3 reduction	Percent salivary CA 15–3 reduction
Cyclophosphamide, methotrexate, fluorouracil	10	No change	49	No change	No change
Paclitaxel	8	11	11	11	No change
Doxorubicin, paclitaxel	1	26	33	No change	No change
Tamoxifen	1	11	80	12	No change
Cyclophosphamide, doxorubicin, fluorouracil	1	No change	20	No change	No change

The *n*-value totals in these tables are more than the population *n*-values due to the fact that some subjects received multiple treatment regimens.

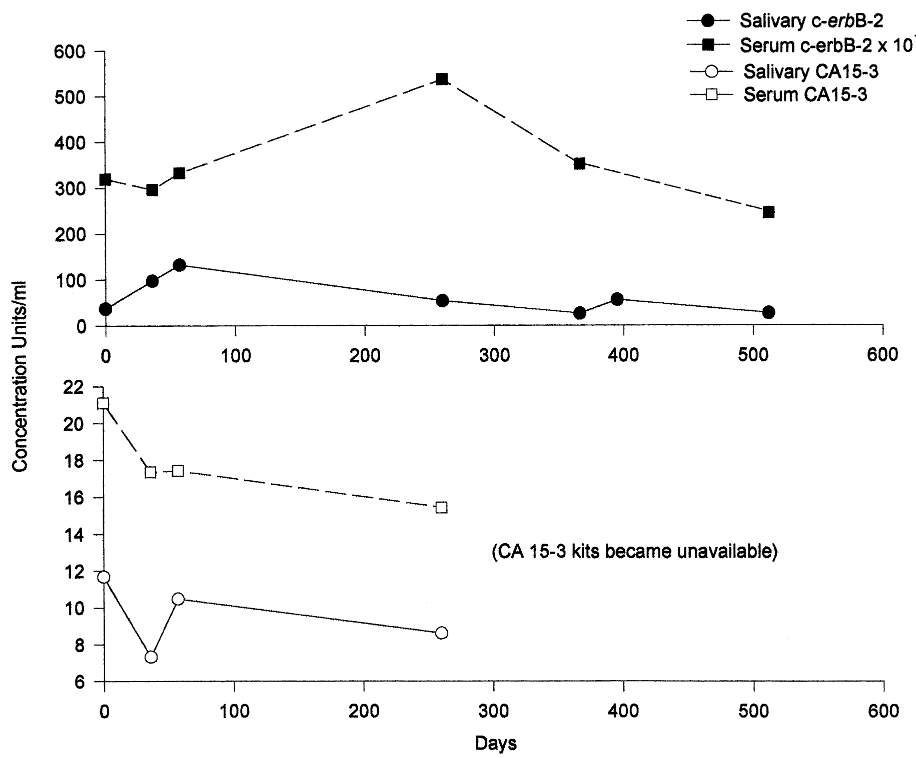

Fig. 1. A 42-year-old Caucasian woman with Stage I, infiltrating ductal carcinoma. She had a grade 2 tumor that was er/pr positive and Her-2 receptor negative. Received only surgical treatment.

Individual patient monitoring is illustrated in Figs. 1–6. As shown in these figures, varying degrees of compliance and responsiveness to therapy were observed.

Discussion

Results of this study suggest that saliva may have a role in monitoring and in the management of breast cancer during the course of the disease. Similar to serum c-*erb*B-2, it has the potential for monitoring patients in both early (24) and late stages of disease (34). Table 1 illustrates the mean values for each marker over time and for each type of treatment regimen. This data, when analyzed using GLM, demonstrated that only two models were statistically significant (Table 2). This was the serum and the salivary c-*erb*B-2 marker models. The serum marker showed significant differences according to treatment mode with serum c-*erb*B-2 concentrations being higher than the other treatment modalities. This is probably due to the fact that the patients receiving a combination of chemotherapy and radiation had a higher percentage (60%) of

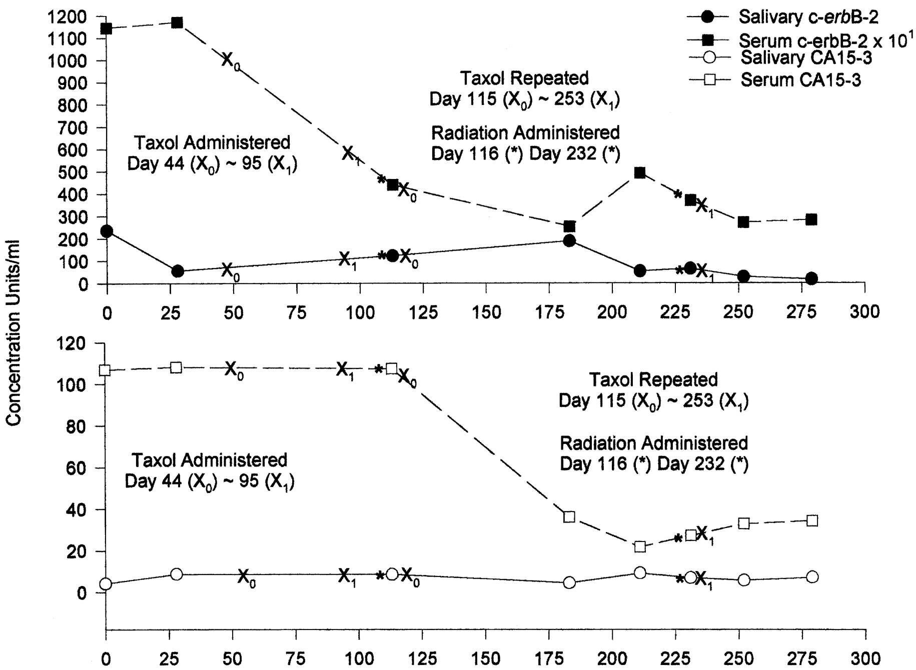

Fig. 2. A 60-year-old African–American woman with Stage IIIa, infiltrating ductal carcinoma. She had a grade 3 tumor that was er/pr and Her-2 receptor positive. Received chemotherapy using paclitaxel as the agent of choice and two doses of radiotherapy.

Her-2/neu-positive subjects than the other treatment regimens. Reports in the literature (21–24) support this finding as serum c-*erb*B-2 assays can better assess these patients than Her-2/neu-negative patients. Salivary c-*erb*B-2, salivary CA 15–3, and serum CA 15–3 did not reflect this finding.

The other GLM analysis that was significant was the salivary c-*erb*B-2 model (Table 2). This model demonstrated a significant decrease in salivary c-*erb*B-2 concentrations with time. This finding may be contributed to fact that salivary c-*erb*B-2 concentrations appear to be independent of Her-2/neu receptor status. The

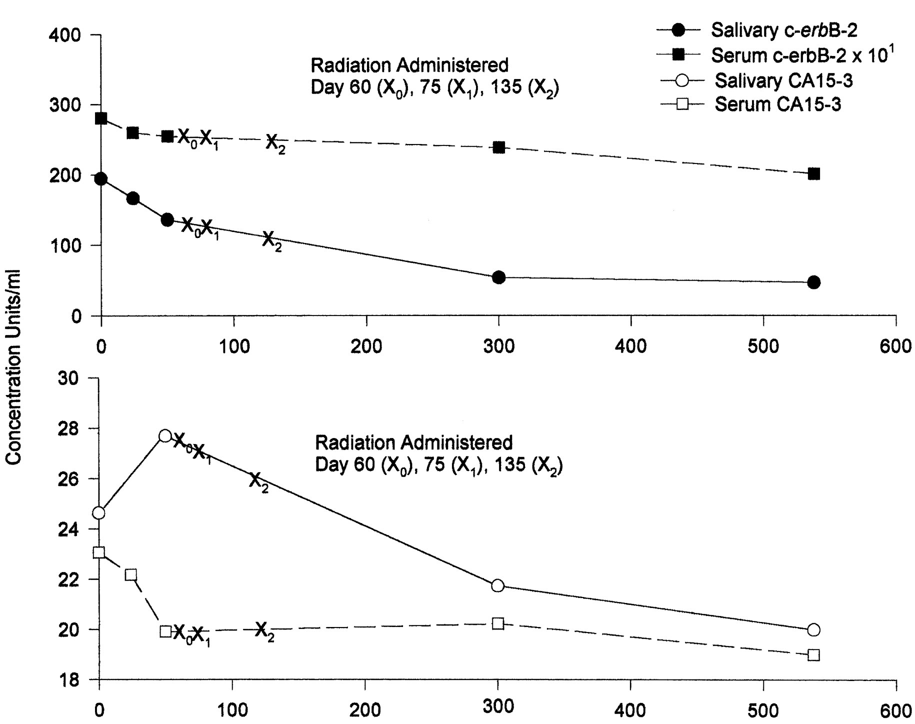

Fig. 3. A 62-year-old Caucasian woman with Stage I, infiltrating ductal carcinoma. She had a grade 1 tumor that was er/pr receptor positive. Her-2 receptor status was unavailable at the time patient entered the study. Received only radiotherapy treatment.

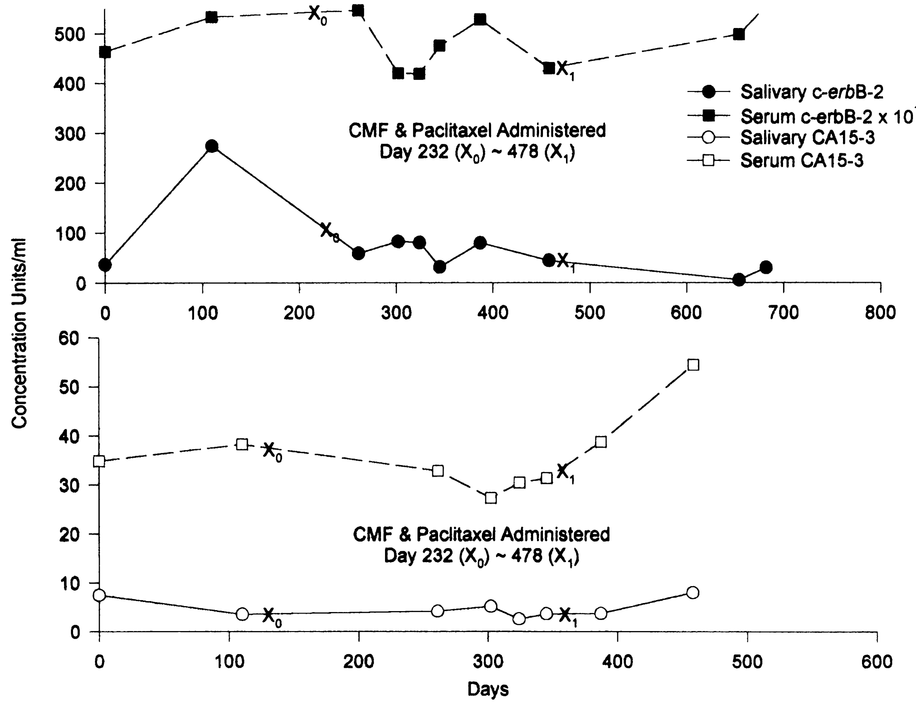

Fig. 4. A 55-year-old Caucasian woman with Stage IIIa, infiltrating ductal carcinoma. She had a grade 3 tumor that was er/pr and Her-2 receptor negative. Received chemotherapy using CMF and paclitaxel.

investigators cannot explain this finding at this time, but are currently conducting research to answer this question. One possibility is that the salivary c-*erb*B-2 protein is not exactly the same as the p185 soluble c-*erb*B-2 protein found in serum. Research using mass spectrometry and protein sequencing are being conducted to determine the nature of this salivary protein that reacts to soluble serum c-*erb*B-2 protein antibodies.

A number of issues were addressed in this study with respect to specific treatment regimens. Table 4 illustrates that only salivary c-*erb*B-2 protein exhibited a significant ($P < 0.001$) pre- and

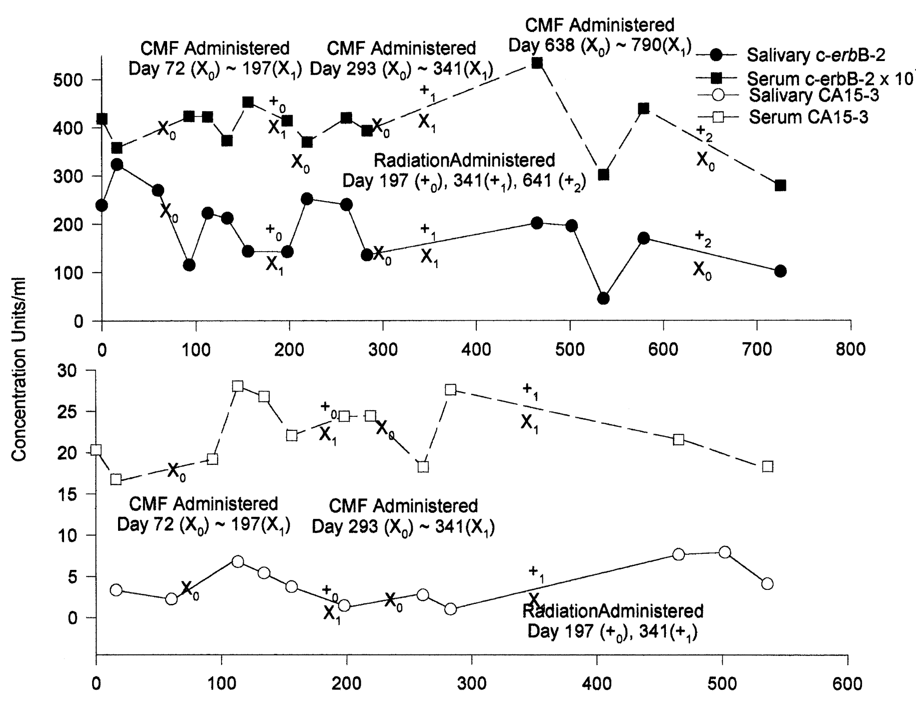

Fig. 5. A 54-year-old Caucasian woman with Stage IIb, infiltrating ductal carcinoma. She had a grade 3 tumor that was er positive and pr receptor negative. Her-2 receptor status was unavailable at the time patient entered the study. Received chemotherapy using CMF as the agent of choice followed by three doses of radiotherapy.

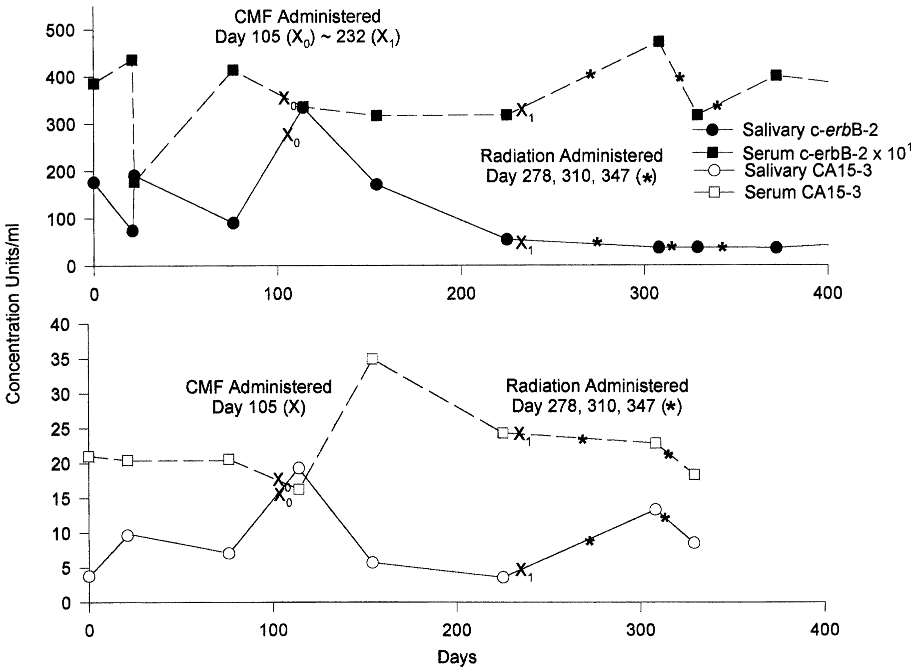

Fig. 6. A 59-year-old African–American woman with Stage IV, infiltrating ductal carcinoma. She had a grade 3 tumor that was er positive and pr receptor negative and Her-2 receptor positive. Received chemotherapy using CMF as the agent of choice followed by three doses of radiotherapy.

post-operative concentration difference. As previously mentioned, not all the tumors were Her-2/neu receptor positive and may explain the findings with respect to serum c-*erb*B-2. Serum and salivary CA 15–3 did not exhibit significant differences between pre- and post-operative marker concentrations, but did demonstrate a decrease in concentrations.

Furthermore, salivary c-*erb*B-2 demonstrated the largest percent change among the group of markers with radiotherapy eliciting the largest percent change in marker concentration. Also, the percentage of assessments below the cut point for salivary c-erbB-2 appears to be comparable to serum CA 15–3 which is used routinely in post-operative monitoring.

Finally, salivary c-*erb*B-2 demonstrated the largest percent change among the group of marker with CMF eliciting the largest percent change in marker concentration.

Figures 1–6 assess marker concentrations over time on a case-by-case basis. Figures 1–6 represent six patients on varying types of cancer therapy with the patient in Fig. 1 receiving only surgery. These patients also vary in age, race, tumor staging, grade and receptor status. The point of interest is that all the marker values mirror each other and are modulated by surgical intervention or by the onset of therapy. These data demonstrate the clinical relevance concerning the use of saliva as a diagnostic medium.

Further research is required to determine the utility of salivary c-*erb*B-2 as a therapeutic monitoring diagnostic. A larger study with more subjects will be required to determine the efficacy of this monitoring technique. Additionally, a larger study can provide information with respect to patient survival, prognosis, and treatment effectiveness. Likewise, the use of these markers in combination and/or in different diagnostic medias warrants further investigation.

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